Explainable Machine Learning: Interpreting Decision Tree Predictions Using Feature Importance and SHAP Values

GitHub: https://github.com/rishivarma1203/Machine-Learning

Student Name: Rishi Varma

Introduction

Machine learning models are used more in practical applications, like healthcare, finance, transport and public services. Numerous machine learning algorithms can make quite correct predictions; however, it can be quite difficult to understand how they do it. Users may find it difficult to put faith in or explain decision-making by the model because of this.

Tools that make machine learning models' predictions more transparent are being developed under Explainable Artificial Intelligence (XAI) to address the problem of black-box models. Explainable models provide transparency to enable practitioners evaluate whether models are dependable, fair and meaningful.

Decision trees are particularly useful for researching explainable machine learning (or interpretable machine learning) because they can be visualized and easily interpreted. Nonetheless, when models grow more complicated, feature importance and SHAP (Shapley Additive Explanations) can expose how its various input variables contribute to its output predictions.

In this tutorial, we will see how explainable machine learning techniques can help us interpret a Decision Tree classifier on the Titanic dataset. The experiment looks at which features are important for survival predictions and how they contribute to model predictions.

Decision Trees and Model Interpretability

Decision Trees are supervised learning algorithms that are often used for classification and regression tasks. The model does this by continuously splitting the data set into sub-data sets based on feature values. When a split is created, it generates a decision rule that divides the data into groups that are similar concerning the target entity.

A typical decision tree consists of three main components:

- Root node, standing for the first decision point.
- Internal nodes, standing for decision rules based on feature values.
- Leaf nodes, standing for final prediction outcomes.

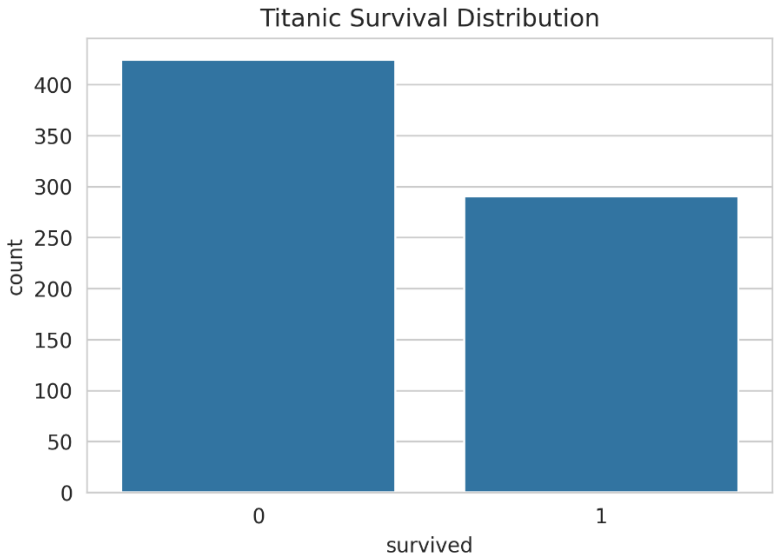
Because the decision process can be visualized as a tree structure, decision trees are considered one of the most interpretable machine learning algorithms. Each branch of the tree corresponds to a rule that leads to a specific prediction.

However, as models become more complex, it becomes increasingly important to analyse which features influence the predictions. Techniques such as feature importance and SHAP explanations help offer deeper insight into how models behave.

Dataset Description

This tutorial uses the Titanic passenger dataset, which holds information about the passengers who boarded the Titanic, such as their demographics and travels. The available attributes of the passenger are used to predict if he survived or not in this machine learning task.

Information on the passengers is described in a number of features. These include the class they travelled in, sex, age and fare. The variable shows whether the passenger survived.



The dataset shows that the number of passengers who did not survive was greater than the number who survived, which creates an imbalanced classification problem.

Model Training

A Decision Tree classifier was trained using the selected features. The dataset was divided into training and testing sets to evaluate model performance.

The model was trained with a limited tree depth to avoid excessive complexity while still capturing meaningful patterns within the dataset. After training the model, predictions were generated for the testing data, and several evaluation metrics were calculated.

These metrics include:

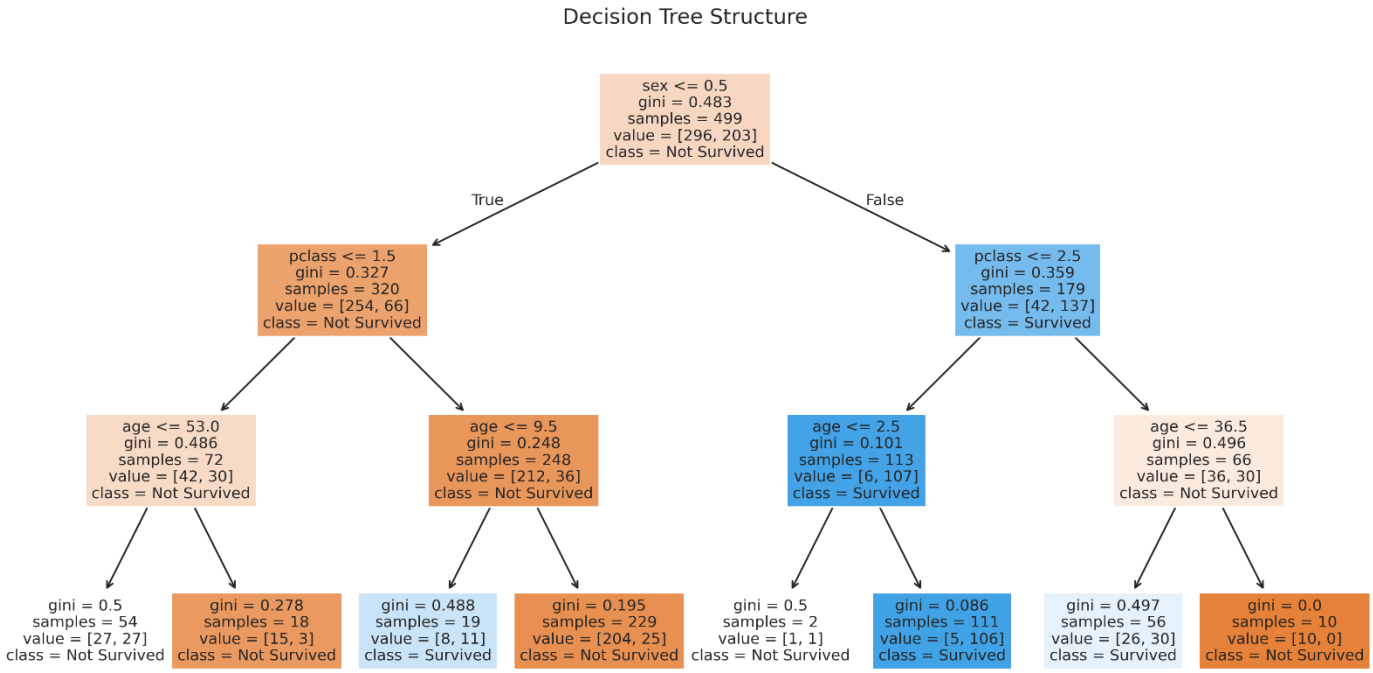
- Accuracy
- Precision
- Recall
- F1 Score

These performance metrics provide an overview of how effectively the model predicts passenger survival.

	Metric	Value
0	Accuracy	0.776744
1	Precision	0.719101
2	Recall	0.735632
3	F1	0.727273

Decision Tree Visualization

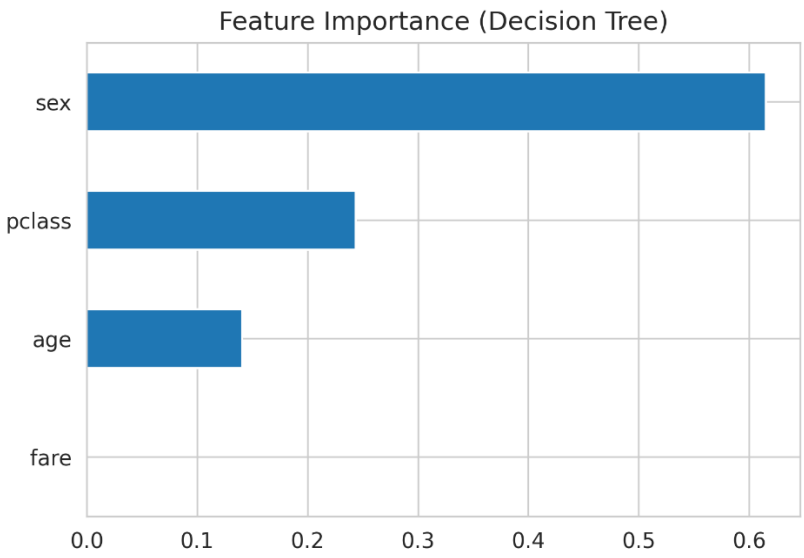
One advantage of decision tree models is that the structure of the model can be visualized directly. This visualization helps illustrate how the model makes predictions based on feature values.



A decision tree is a model which predict the value of target variable y. When you follow the branches of the tree, it shows how the model arrived at that prediction. The decision tree visualization illustrates the influence of gender and passenger class on predictions.

Feature Importance Analysis

Feature importance measures the relative influence of each input variable in the model's decision-making process. Decision tree algorithms calculate feature importance by evaluating how much each feature contributes to reducing classification error during tree construction.



The feature importance plot reveals which variables have the greatest influence on the survival prediction. In many Titanic analyses, gender and passenger class are often among the most important predictors.

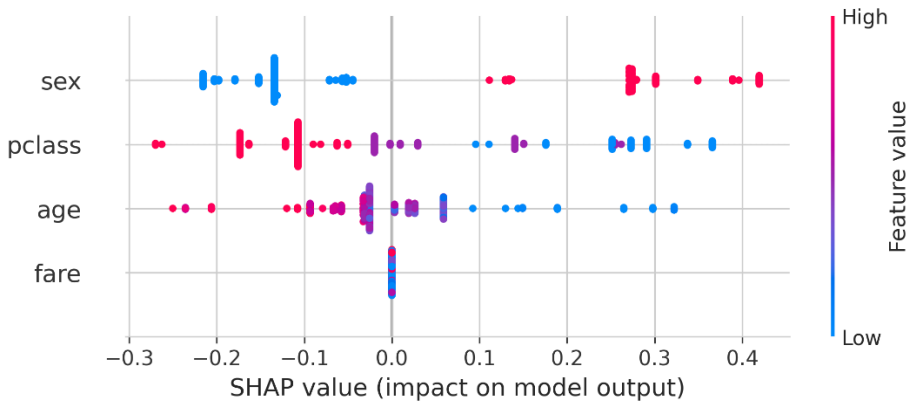
Understanding feature importance helps analysts find which factors are most influential when the model makes predictions.

SHAP Explanations

While feature importance provides a global overview of feature influence, SHAP values offer a more detailed explanation of model predictions.

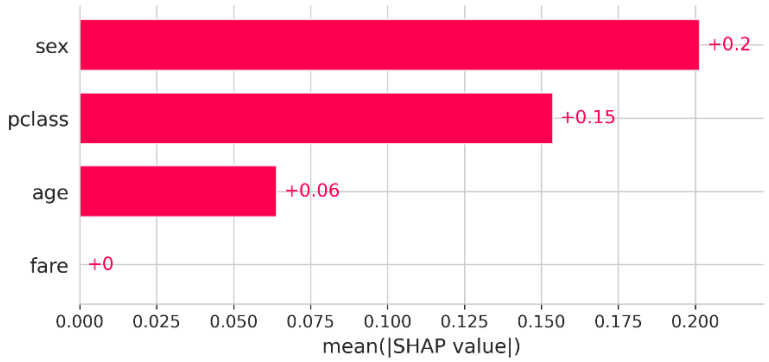
SHAP values are based on concepts from cooperative game theory and measure how each feature contributes to the prediction outcome. Each feature receives a SHAP value showing whether it increases or decreases the predicted probability of survival.

The SHAP summary plot provides a global explanation of feature influence across the entire dataset.



Every point here is an observation in the data. The placement of each point denotes whether the variable raises or lowers the predicted probability.

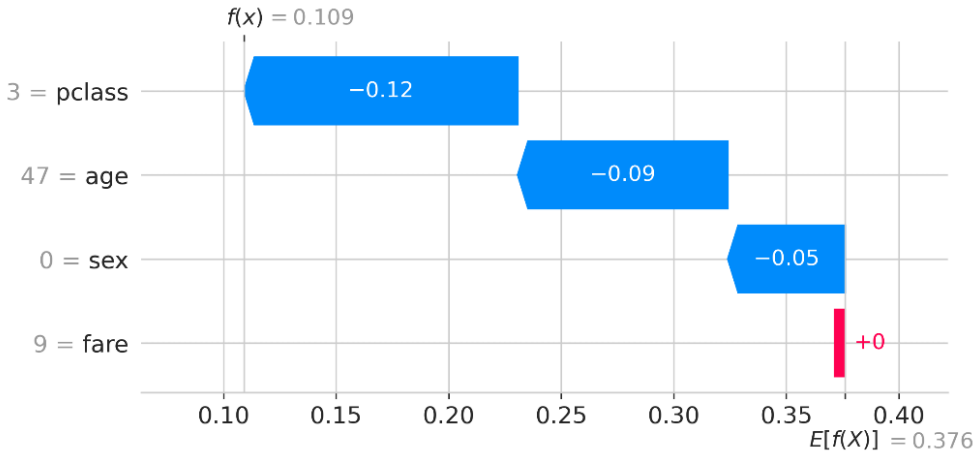
A SHAP bar plot depicts the global feature importance.



These plots help illustrate how unique features influence predictions across the dataset.

Individual Prediction Explanation

SHAP has a major advantage in explaining how individual predictions are arrived at. The waterfall plot proves the contribution of each feature to the final prediction for a passenger.



A positive contribution increases the predicted probability of survival whilst a negative one decreases it. By looking at these contributions, users can understand how the model explains its prediction.

In situations where sensitive decisions are made, this type of explanation is extremely useful.

Discussion

By the end of this tutorial, we will see how explainable machine learning techniques can help us understand predictive models. Machine learning models are often evaluated on their prediction accuracy. However, how the model comes to such predictions is equally important when the model is being used for decisions changing humans.

The visualization of the decision tree gives a clear picture of how the model is splitting up passengers. By analysing the decision rules in the tree, one can understand which variables are most important to the model making a prediction. The model can first split the data based on gender or passenger class, which are patterns that are very strongly associated with a person's chances of survival on the Titanic.

Using feature importance analysis, we can also make sure that how much each of the variables affects the prediction. Some variables have more predictive ability than others when deciding survival probability, according to results. In numerous instances, the gender and passenger class variables appear as some of the most impactful, showing they hold predictive power in the dataset.

Nonetheless, only a global interpretation of the model comes out of feature importance. It shows which attributes are generally important, but not how they influence a particular prediction. SHAP explanations offer more information in this area. The SHAP summary view illustrates the effect of different feature values for each passenger on the predictions. For instance, a particular value of a feature can increase the probability of survival and vice versa.

The SHAP waterfall plot adds even more detail to the explanation by showing how the prediction of a specific feature was achieved. Through analysing these attributions, one can discern why the model predicted a specific outcome for a particular passenger. This type of explanation increases transparency and enables users to better understand machine learning decisions.

Overall, looking at the decision trees, the feature importance and the SHAP explanations gives us good insights into the model behaviour. The mechanisms illustrate how explainable AI (XAI) techniques can close the gap between prediction accuracy and transparency.

Limitations

The lesson will show how explainable machine learning techniques can help improve model interpretability, but there are limitations.

Initially, a small subset of features was used from the Titanic dataset for the analysis. While the chosen variables give essential information about passengers' survivals, more features can increase the prediction performance of the model. Adding more variables may also alter the relative importance of the variables found in our study.

In the second phase, we used a single decision tree model with limited tree depth to keep interpretability. Although this tactic helps in supporting a simple model structure that is easier to understand, it may affect predictive performance compared to sophisticated machine learning models.

The set upon which this study is done has certain drawbacks. Historical events such as the Titanic dataset have limited observations. Hence, the patterns learnt by the model may not be transferable to other classification problems. Consequently, the results should be viewed more as an educational case than as a generalizable predictive model.

In the end, the SHAP explanations help you understand the model better but still need interpretation. Misunderstanding feature contributions may result in misleading conclusions about the actual data relationships.

Regardless of these limitations, the tutorial shows how it is possible to use explanatory machine learning to improve transparency and understanding.

Future Work

This analysis could be extended to study more machine-learning models and compare their interpretability. Ensemble techniques like Random Forest or Gradient Boosting models typically perform better than stand-alone counterparts, although they lack interpretability. But a trade-off is needed.

One possible extension would involve working with larger and more complex datasets using explainable machine learning. Real-world applications tend to have many more features and observations, which can complicate the interpretation of the model.

To further investigate how machine learning models react to input variables, future studies could investigate alternative explainability techniques LIME (local interpretable model agnostic explanations) and partial dependence plots.

Conclusion

Through this tutorial, you will be learning how to make use of explainable machine learning techniques to interpret the workings of a decision tree classifier. By using the Titanic dataset, the experiment showed feature importance and SHAP explanations provide insights into what influences model predictions.

Through the construction of decision trees, we were able to assess the risk to a beneficiary across distinct levels of insurance. SHAP analysis made it clear how to interpret the result to decide risk. These tools enhance the transparency and trustworthiness of ML models.

To deploy machine learning safely in practice, it is important to find out how models predict. Through AI techniques we can make sure that predictive models stay interpretable and accountable.

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